[PHYS 6268] Nonlinear Dynamics: Problem Set 13

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Strogatz Book: Problems 10.5.2, 10.5.7, 11.3.3, 11.3.5, 11.4.6, & 12.2.18

1 Problem 10.5.2: Calculate the Liapunov Exponent for the decimal shift map $x_{n+1} = 10x_n \mod 1$

The Liapunov exponent λ_L represents the eigenvalue along a repelling dimension in phase space that characterizes the separation between two arbitrarily close initial conditions, separated by δ_0, that is, $(x_0, x_0 + \delta_0)$. Intuitively, this eigenvalue multiplier quantifies the sensitivity to initial conditions admitted by the governing system. The Liapunov spectrum, in general, has a size corresponding to the dimension of the system, which here is only a single dimension. Hence, we merely consider the unit interval (defining the one dimensional phase space of the decimal shift map) $[0, 1]$. If we consider the distance of separation δ_n as a function of number of map iterations n (where $f(x_n) = x_{n+1}$), then the Liapunov exponent can be written as

$$\lambda_L = \lim_{n\longrightarrow\infty} \frac{1}{n} \ln\left|\frac{\delta_n}{\delta_0}\right| = \lim_{n\longrightarrow\infty} \frac{1}{n} \ln\left|\prod_{i=0}^{n-1} f'(x_i)\right| = \lim_{n\longrightarrow\infty} \left(\frac{1}{n}\sum_{i=0}^{n-1} \ln|f'(x_i)|\right) \quad . \tag{1}$$

Therefore, we must evaluate the limit $\lim_{n\longrightarrow\infty}\left(\frac{1}{n}\sum_{i=0}^{n-1}\ln|f'(x_i)|\right)$ for the decimal shift map

$$x_{n+1} = f(x_n) = 10x_n \mod 1 \quad .$$

The form of f is linear in x_n, which can be verified by sketching the portrait of x_n versus x_{n+1}: ten consecutive linear curves are arranged in sequence, discontinuous at each decimal tenth $(0.1, 0.2, \ldots, 0.9)$ where the value of x_{n+1} falls from $0.\bar{9}$ to 0 due to the modulo operator. However, even though the map itself is discontinuous due to the $\mod 1$ factor, the *slope* of the mapping remains a constant value of 10.

Therefore, we know that *regardless* the choice of x_0, for an arbitrary value of x_i after some $i < n$ iterations of f applied to x_0, the value of $f'(x_i) = 10$ *for all* $x \in [0, 1]$. Therefore, equation (1) for the Liapunov exponent becomes

$$\begin{aligned}
\lambda_L &= \lim_{n\longrightarrow\infty}\left(\frac{1}{n}\sum_{i=0}^{n-1}\ln|f'(x_i)|\right)\\
&= \lim_{n\longrightarrow\infty}\left(\frac{1}{n}\sum_{i=0}^{n-1}\ln|10|\right)\\
&= \lim_{n\longrightarrow\infty}\left(\frac{1}{n}\cdot n\ln|10|\right)\\
&\implies \lambda_L = \ln 10 \quad .
\end{aligned}$$

2 Problem 10.5.7: Figure 10.5.2 (Strogatz) suggests that $\lambda = 0$ at each period-doubling bifurcation value r_n. Show analytically that this is correct.

In order to show this result analytically, we must utilize the definition of the Liapunov exponent in equation (1) from Problem 1. We will use information about the period-doubling phenomenon to evaluate this expression. It is important to note that a period-doubling bifurcationis associated with a flip bifurcation, with supercritical indicating a stable cycle exists at greater values of x than the fixed point, and subcritical indicating an unstable cycle exists below the bifurcation. In the case of the map exhibiting period-doubling here, the newly-birthed cycles are stable.

We will build off of this to determine the value of λ_L at the critical points of period-doubling. At the moment of period doubling, there exists a stable, newly "birthed" 2-cycle around the fixed point. At the location of the fixed point during a flip bifurcation, the derivative of the map $f'(x^*) = -1$. Hence, we have that $f'(x_i) \approx -1$ for all points x_i visited by the trajectory on the 2-cycle, since the cycle exists in the neighborhood of the fixed point. We let the first point on the cycle be denoted by x_C. Since the cycle is stable, all points in the future, that is, x_j for $j > C$ are *also* on the cycle, and *also* in the neighborhood of the fixed point where $f'(x_j) \approx -1$. Then, we can write the following, with C being the (arbitrarily large) *first* value of the mapping that reaches the cycle:

$$\lambda_L = \lim_{n\longrightarrow\infty} \left(\frac{1}{n} \sum_{i=0}^{n-1} \ln|f'(x_i)| \right) \approx \lim_{n\longrightarrow\infty} \frac{1}{n} \left(\sum_{i=0}^{C-1} \ln|f'(x_i)| + \sum_{i=C}^{n-1} \ln|-1| \right) \quad .$$

Here, we have separated the series in the Liapunov exponent definition between the subset of the trajectory that approaches the cycle and the subset actually on the cycle. Since $\ln|-1| = 0$, the expression becomes

$$\lambda_L \approx \lim_{n\longrightarrow\infty} \frac{1}{n} \left(\sum_{i=0}^{C-1} \ln|f'(x_i)| \right) \quad .$$

We know that $f'(x_i)$ remains finite for all values of x_i. Let A denote the maximum value of the series, such that

$$A = \max\{\ln|f'(x_0)|, \ln|f'(x_1)|, \ldots, \ln|f'(x_{C-1})|\} \quad .$$

Then, we know that the series

$$\sum_{i=0}^{C-1} \ln|f'(x_i)| \leq \sum_{i=0}^{C-1} A = A(C-1) \qquad ,$$

that is, the value of the series *must* be finite (though it may be very, very large). However, if the value of the series is finite, then it must vanish when divided by infinity as the limit is taken. Formally,

$$\lambda_L \approx \lim_{n\longrightarrow\infty} \frac{1}{n} \left(\sum_{i=0}^{C-1} \ln|f'(x_i)| \right) \leq \lim_{n\longrightarrow\infty} \frac{1}{n} \left(\sum_{i=0}^{C-1} A \right) \qquad ,$$

which yields the limit

$$\lambda_L \approx \lim_{n\longrightarrow\infty} \frac{1}{n} \left(\sum_{i=0}^{C-1} \ln|f'(x_i)| \right) \leq \lim_{n\longrightarrow\infty} \frac{A(C-1)}{n} = 0 \qquad ,$$

for finite A, C. Therefore, at the instant of period-doubling due to the flip bifurcation, the Liapunov exponent $\lambda_L = 0$.

However, this rationale only applies at the *first* instance of period doubling, where the flip bifurcation occurs and the fixed point produces the 2-cycle. In order to consider the behavior of this map at a general stage of period doubling, we consider a point p on the stable 2-cycle. Then, p is a fixed point of the map $f(f(p)) \equiv f^2(p) = p$. In general, for a $2n-$cycle, the points q that compose the cycle are fixed points of the map iterated $2n$ times: $f^{2n}(q) = q$. We must then use information about the specific map to determine the Liapunov exponent for *any* period-doubling scenario, not just at the bifurcation.

$$f(x_i) = rx_i(1-x_i)$$

defines the map. Hence, the bi-iterated composition of the map reads

$$\begin{aligned} f(f(x) &= r(rx(1-x))(1-(rx(1-x))) \\ &= r^x(1-x) - r^2x^2(1-x)^2 \\ &= r^2x - r^2x^2 - r^2x^2 + 2r^2x^3 - r^2x^4 \\ &= r^2(x - 2x^2 + 2x^3 - x^4) \quad . \end{aligned}$$

Hence, points on the 2-cycle are roots of the above expression, representing fixed points of the bi-iterate map $f \circ f \equiv f(f(x))$.

Then, if we know our point to be a fixed point of $f \circ f(x)$, we have that

$$f(f(x) = x \implies (f^2)'(x_0) = 1 \qquad .$$

Differentiating f^2, we find

$$r^2(1 - 2x + 6x^2 - 4x^3) \qquad .$$

Now, it is important to utilize the result from Strogatz that says for a $2n-$cycle, the Liapunov exponent is given by

$$\lambda_L = \frac{1}{2n} \sum_{i=1}^{2n-1} \ln |f'(x_i)| \qquad .$$

Thus, using the chain rule, this implies that

$$\lambda_L = \frac{1}{2n} \sum_{i=1}^{2n-1} \ln |f'(x_i)| = \frac{1}{2n} \ln \left|(f^{2n})'(x_0)\right| \qquad .$$

Therefore, using the result for the derivative at the instant period-doubling, we now have that

$$\lambda_L = \frac{1}{2n} \ln \left|(f^{2n})'(x_0)\right| = \frac{1}{2n} \ln |1| = 0 \qquad .$$

3 Problem 11.3.3: Generalization of the even-fifths Cantor set

The "even–sevenths Cantor set" is constructed as follows. Divide the unit interval $[0, 1]$ into seven equal pieces; delete pieces 2, 4, and 6; repeat the process on sub-intervals recursively.

3.1 Part (a): Find the similarity dimension of the set.

The similarity dimension for a self-similar set represents the quantitative relationship between successively scaled subsets.[1] For a set composed of m copies of itself, scaled by a factor of r, the similarity dimension is defined as

$$d \equiv \frac{\ln m}{\ln r} \qquad .$$

With this definition, it is evident that the approach to determine the similarity dimension requires computing a few iterations of the self-similarity and comparing them.

Before we calculate the similarity dimension for the even-sevenths Cantor set, we review the analogous calculation for the middle-thirds Cantor set. This will simplify the process of generalization solicited in section 3.2. Figure 1 depicts a geometric representation of the Cantor middle-thirds set using line segments. The subscript of the set S_n represents the nth iteration of the Cantor process.

[1] It should be noted that we will refer to the "size" of sets rather loosely here, and we mean by "size of sets" the fraction of the *length* of the unit interval (or appropriate domain) occupied by the closed set in question.

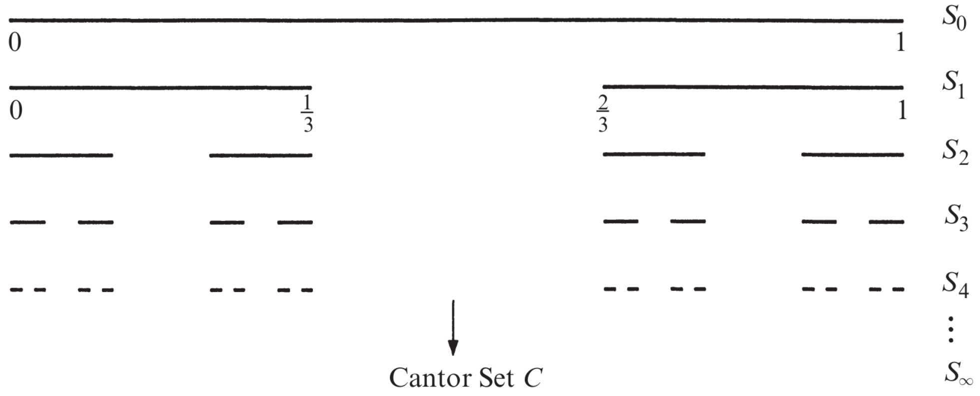

Figure 1: Middle-thirds Cantor set construction. Adopted from Figure 11.2.1 of Strogatz.

By inspection of Figure 1, the set is scaled by a factor of $r = 3$ in each successive iteration, i.e., the set is reduced in size by a factor of 3 when incrementing from S_k to S_{k+1} for some $k \in \mathbb{N}$. In addition, the domain of the scaled subset, compared to the preceding set, is one half. Hence, the set reduction factor is $m = 2$ The similarity dimension of the middle-thirds Cantor set is thus given by

$$d_3 = \frac{\ln 2}{\ln 3} \approx 0.63093 \qquad .$$

This approach can now be readily extended to the even-sevenths Cantor set. In contrast to the construction in Figure 1, now we are subdividing each set into 7 intervals, and dropping the middle "even" sets free. Then, the interval size at each successive step is 1/7 the size of the set in the previous iteration. Therefore, for this Cantor set, $r = 7$. Correspondingly, the domain of the set is divided more finely amongst self-similar sets each iteration. Since the interval is divided into seven subsets, and four are "kept", the similarity is represented on scales equal to a quarter the size of the original set. That is, $m = 4$. Therefore, the similarity dimension for the even-sevenths Cantor set is

$$d_7 = \frac{\ln 4}{\ln 7} \approx 0.71241 \qquad .$$

3.2 Part (b): Generalize the construction to any odd number of pieces, with the even ones deleted. Find the similarity dimension of this generalized Cantor set.

We define the Cantor even-N set to be the generalized Cantor construction: the unit interval is divided into N pieces, and the inner even pieces are deleted. Thus, N is odd; the number of pieces retained is $N/2 + 1/2$, and the number of sets deleted is $N/2 - 1/2$. By analogy to the middle-thirds, even-fifths, and even-sevenths Cantor sets, this set exhibits self-similar behavior comparing successive iterations.

Using this definition, we can immediately extend the above calculation of similarity dimension to the generalized Cantor even-N set. The scaling of each self-similar segment corresponds to a factor of N: the unit interval is divided, by definition, into N pieces. Hence, $r = N$. This agrees with our values of $r = 3$ and $r = 7$ above. Moreover, the number of segments *retained* corresponds to the division of the set domain. Since the even natural number corresponding to $N/2 + 1/2$ segments are retained, the set domain is divided by a factor of $N/2 + 1/2$. Thus, $m = (N + 1)/2$.

Substituting these quantities into the similarity dimension of the generalized Cantor even-N set, we arrive at

$$d_N = \frac{\ln\left(\frac{N+1}{2}\right)}{\ln N} = \log_N(N+1) - \log_N(2) \qquad .$$

By considering the limit of this expression as it goes to infinity, to leading order it will follow the lowest order term of the series expansion at $N = \infty$,

$$d_N \approx \frac{\ln N - \ln 2}{\ln N}$$

when N is very large. Hence, as we let N approach ∞, we find that

$$\lim_{N \longrightarrow \infty} d_N = 1 \qquad .$$

4 Problem 11.3.5: Find the similarity dimension of the unit interval subset consisting of real numbers that can be written without the digit 8 appearing anywhere in their decimal expansion.

In order to find the similarity dimension, we need to construct an iterative approach that tends towards the set described above. The elements of our set $x_i \in S$ can be written (in decimal form) as

$$x_i = 0.x_{i,1}x_{i,2}x_{i,3}\ldots x_{i,N} \qquad .$$

For rational numbers, either N is finite or there is a periodicity in $x_{i,k}$ for some $k \in \mathbb{N}$. For irrational numbers N vanishes, of course. In either case, we can think about our iterated set in the sense of this expansion: this relies on the fact that $x_{i,k}$ are natural numbers between 0 and 9. We start with a representation of all numbers on the unit interval. Then, at the first instance, we restrict $x_{i,1}$ to the set $[0, 1, \ldots, 7] \cup [9]$. This corresponds to eliminating all numbers with a leading decimal of 8. But this only accounts for a large (but finite) subset of the total numbers in question. Now, we advance to the second stage: restrict $x_{i,2}$ to the set $[0, 1, \ldots, 7] \cup [9]$.

This process must be repeated infinitely in order to arrive at the final set S, since irrational numbers have an infinite aperiodic decimal expansion and are dense on the unit interval. However, similar to the Cantor sets in Problem 3, we can compare successive iterations of the construction that *approach* the set. By the self-similarity property (evidently a feature of this set construction and the others above), each successive iteration will yield the *same* values for m and r that are used to calculate the similarity dimension. Thanks to the fractal and self-similar properties of this set, we can use the same approach as above to determine the similarity dimension d.

Since the iterative procedure of the set is continuously focusing in on lower and lower orders of magnitude (i.e., digits grow farther and farther from the leading one in the decimal expansion), the self-similar scaling of the set occurs on a scale of $r = 10$. That is, after removing the value $x_{i,2} = 8$ from the set, there will be separate sets on the sub-intervals $[0.0, 0.1), [0.1, 0.2), \ldots$, each occupying one-tenth the space of the set created by the removal of $x_{i,1} = 8$.

Since one digit of the possible 10 is removed, there are nine remaining "stable" values of $x_{i,k}$ at each k. Hence, the set is composed of nine self-similar "copies" of itself, and the value of $m = 9$. Therefore, the similarity dimension for the unit interval subset consisting of real numbers that can be written without the digit 8 appearing anywhere in their decimal expansion is given by

$$d = \frac{\ln m}{\ln r} = \frac{\ln 9}{\ln 10} \approx 0.95424 \qquad .$$

It makes intuitive sense that the similarity dimension of this fractal set on the interval would be larger than that of, e.g., the middle-third or even-fifth Cantor sets, where a much larger fraction of the set is removed each iteration.

5 Problem 11.4.6: A strange repeller for the tent map

The tent map on the unit interval $[0,1]$ is defined by $x_{n+1} = f(x_n)$, where the mapping

$$f(x) = \begin{cases} rx \quad ; & 0 \le x \le \frac{1}{2} \\ r(1-x); & \frac{1}{2} \le x \le 1\,. \end{cases} \tag{2}$$

and $r > 0$. Here, we will assume $r > 2$. Then some points get mapped outside the unit interval. If $f(x_0) > 1$, then we say that x_0 has "escaped" after one iteration. Similarly, if $f^n(x_0) > 1$ for some finite integer n, but $f^k(x_0) \in [0,1] \quad \forall k < n$, then we say that x_0 has escaped after n iterations.

5.1 Part (a): Find the set of initial conditions that escape after one or two iterations.

To do so, we must begin by evaluating which values are mapped out of the unit interval in a single iteration. Then, we shall expand this to two mappings by considering the pre-Image of the set mapped out of the domain $I = [0,1]$. Thus, we consider elements from the lower half interval $I_L \equiv [0, \frac{1}{2})$ that are cast beyond unity or those in the upper half interval $I_U \equiv [\frac{1}{2}, 1]$ cast beyond it as well.

We begin with the map in the upper half. On the domain I_U, the term in the map $(1-x)$ ranges from 0 to 1/2, creating the symmetry of the "tent". For a value to be cast out, it must fall in the range

$$\left(\frac{1}{2}, 1 - \frac{1}{r}\right) \qquad .$$

On the other hand, values in the range

$$\left(1 - \frac{1}{r}, 1\right)$$

remain in the set.

For the lower interval, the "safe" range is $(0, 1/r)$ and the values in the range $(1/r, 1/2)$ are cast beyond unity, out of the interval. Hence, all values in the set

$$\left(\frac{1}{r}, \frac{1}{2}\right] \bigcup \left(\frac{1}{2}, 1 - \frac{1}{r}\right)$$

are cast out after a single iteration. Now, we consider those cast out after two. Following the outlined strategy, we identify the region

$$\left(\frac{1}{r^2}, \frac{1}{r} - \frac{1}{r^2}\right) \bigcup \left(1 - \left(\frac{1}{r} - \frac{1}{r^2}\right), 1 - \frac{1}{r^2}\right) \qquad ,$$

where the LHS of the union is a subset of I_L and the RHS is a subset of I_U. Hence, the set of initial conditions x_0 that escapes in 1 or 2 iterations is described by

$$x_0 \in \left(\frac{1}{r^2}, \frac{1}{r} - \frac{1}{r^2}\right) \bigcup \left(\frac{1}{r}, \frac{1}{2}\right] \bigcup \left(\frac{1}{2}, 1 - \frac{1}{r}\right) \bigcup \left(1 - \left(\frac{1}{r} - \frac{1}{r^2}\right), 1 - \frac{1}{r^2}\right) \subset I$$

5.2 Part (b): Describe the set of x_0 that *never* escape.

Consider, for instance, $r = 3$ as an example satisfying $r > 2$. If $x_0 = 1/4$, then $f(x_0) = rx_0 = \frac{3}{4}$. Now consider $f \circ f(x)$. $f(x_0) = 3/4 \implies f \circ f(x_0) = r\left(1 - \frac{3}{4}\right) = 3(1/4) = 3/4$. Since the value 3/4 falls back into the domain of I_U, it is again fed into the mapping $f(x) = r(1-x) = 3/4$. Hence, it is a fixed point $x^* = 3/4$ of the mapping $f_{r=3}$, and never escapes. It also has a pre-Image 1/4.

It is important to notice that only the map for the upper interval I_U possesses a nonzero fixed point. Hence, both this fixed point x^* and the pre-image of this fixed point x_p^* from the lower interval $rx = x^*$ are elements of the set that never escapes. However, 0 is always a fixed point of rx, regardless the value of r. Analogously, so is unity for the mapping in the upper interval. We note that these are included in the unit interval domain of the mapping.

In general, the fixed point is given by

$$x = f(x) = r(1-x) \implies x^* = \frac{r}{1+r} \qquad .$$

Hence, both x^* and its pre-Image

$$x_p^* = \frac{x^*}{r}$$

should it fall in the lower domain. That is, x_p^* is an element of the set that never escapes *only* if

$$\frac{r}{r(1+r)} < \frac{1}{2} \implies$$

which is always satisfied since $r > 1$ by definition.

In order to determine if there are any other nonzero values which never escape (i.e., away from the fixed point and its pre-image) we return to our example with $f_{r=3}$ above and test nearby quantities. Testing $x_0 = 1/5$ yields an escaped value after two iterations. Testing $x_0 = 2/5$ yields an escaped value after two iterations as well. Revisiting the mapping, though, we notice that the value $1/4$ may have a pre-Image from the upper domain and map as well. Sure enough,

$$f(x) = 3(1-x) = \frac{1}{4} \qquad \implies \qquad x = \frac{11}{12} \qquad ,$$

in the upper interval I_U, which similarly has yet another pre-Image $11/36$ from the lower interval. Continuing the chain, we find that $97/108, 97/324, 875/972$ and $875/2916$ also are a part of the set. Thus, it appears the set may not be finite. Therefore, we must generalize this to an expression such that we can describe the set.

Also, the pre-Image of unity $1 - 1/r$ is a part of the set that never escapes. Hence, so far, we have identified the following set that never escapes for the general map:

$$E = \left[0, \frac{1}{1+r}, \frac{r}{1+r}, 1 - \frac{1}{r}, 1\right] \ .$$

Now we must find an expression for the (apparently) indefinite chain of pre-Images for these values. Particularly, we focus on the compounded divisions and multiplications by 3. We ultimately fail to find a general expression for the terms in the sequence, but we note that we can define the set that never escapes as the above set and their pre-Images. Since this set is described by composed, inverted iterations of the mapping

$$g(x) \equiv f^{-1}(x) \qquad ; \qquad g \circ g \circ \ldots g(x) \qquad ,$$

we pose that it is countable (even if possessing an infinite number of elements).

5.3 Part (c): Find the box dimension of the set of x_0 that never escape. This set is called the *Invariant set.*

The box dimension is defined as the size given by the limit

$$D = \lim_{\epsilon \longrightarrow 0} \frac{\log N(\epsilon)}{\log \frac{1}{\epsilon}} \qquad ,$$

where N is defined as the number of boxes needed to cover the set.

Since the domain is a 1-D interval, the "boxes" will be small line segments, that is, finite, closed and proper subsets of the unit interval I. Since there exists a countable sequence of pre-Images to the fixed points described by E, there is also a countably infinite number of infinitely small coverings.

One one hand, since countably infinite subsets of the unit interval I share a 1-1 correspondence with the countably infinite rational numbers, and the rational numbers are dense on the unit interval, seemingly a dense subset of infinitely small line segments would cover the entire unit interval even for infinitely small intervals, that is, as $\epsilon \longrightarrow 0$. This would imply a box dimension of unity. On the other hand, considering the strategy used by Strogatz, if we consider a single, finite iteration of the sequence to evaluate the limiting box dimension, then there are only a finite number of points: which still only require infinitely small segments to cover, implying a box dimension that approaches zero. We are not entirely sure which rationale prevails, but it likely requires a slightly more comprehensive description of the fractal structure of the strange repeller.

5.4 Part (d): Show that the Liapunov exponent is positive at each point in the invariant set.

Since the invariant set remains in the system indefinitely, the average over infinite number of terms will return a finite value in the limit. More precisely, we notice that

$$|f'(x_i)| = r \qquad ,$$

for the Tent map. Thus, the definition of the Liapunov exponent becomes

$$\lambda_L = \lim_{n \longrightarrow \infty} \left(\frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)| \right) = \lim_{n \longrightarrow \infty} \left(\frac{1}{n} \sum_{i=0}^{n-1} \ln r \right) = \ln r \geq \ln 2 > 1 \qquad .$$

Hence, λ_L takes a positive value everywhere in the invariant set.

6 Problem 12.2.18: Show numerically that the Lozi map has a strange attractor when $a = 1.7$ and $b = 0.5$.

The Lozi map is described by

$$x_{n+1} = 1 + y_n - a|x_n| \quad , \qquad y_{n+1} = bx_n \quad ,$$

where $a, b \in \mathbb{R}$ that satisfy $|b| < 1$. This is similar to the Hénon map.

A strange attractor is an attractor that exhibits sensitive dependence to initial conditions. It is a fractal set of fractional dimension that represents an infinite complex of surfaces, extremely close to each other in space. An attractor is defined formally as a *minimal*, *invariant* set that attracts an *open set of initial conditions*. Hence we can show numerically that there is a strange attractor in the map domain by computing a large number $N \gg 1$ of iterations of the map, say $N \longrightarrow 2 \cdot 10^5$, and plotting their arrangement. By inspection of the state space as a whole, the iterated coordinates of the mapping (x_n, y_n) arrange themselves onto a surface that takes some shape. Then, by focusing on successively smaller subsets of the domain and comparing the arrangement of points, the self-similar fractal nature of the strange attractor can be deduced.

This exact process is depicted in Figure 2. By inspection of the top panel, that is, the entire domain of the map, it is evident that the points cluster onto some sort of "folded" object, the strange attractor. This agrees with the similarity and analogy to the Hénon map in Figure 12.2.3 of Strogatz. By comparing the two successively smaller subsets of the domain in the middle and lower panels, inspection indicates that the expected self-similar behavior does indeed occur.

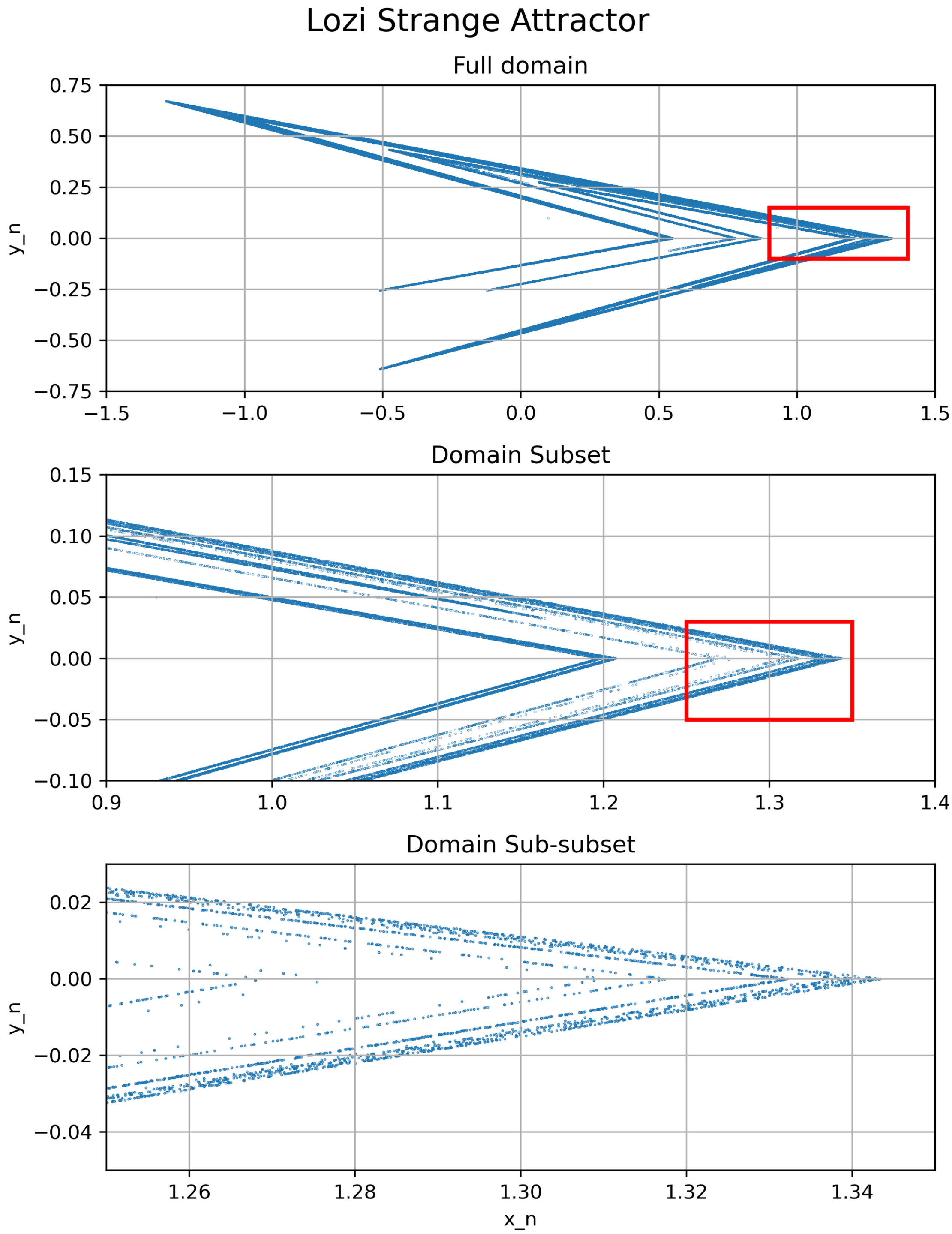

Figure 2: $N = 2 \cdot 10^5$ iterations of the Lozi map (see Problem 6) depicted on three successively smaller domains (moving downward). The red boxes in the upper two panels indicate the subset of the domain shown in the panels below the respective box.

7 Software Usage

The entirety of the work was carried out on pen and paper prior to typesetting, other than plotting and the computation of the Lozi map in Section 6: this were completed using python (`numpy` and `matplotlib`) libraries.

8 Acknowledgments

Thank you both, Alex and Brandon, for your efforts combing my solutions over the past semester.